

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I G. Manobhi Ram- 192311322 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Manobhi Ram

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Unified Big Data Architecture for a Multinational Retail Organization

A multinational retail organization generates huge amounts of data from **billing systems, warehouses, online shopping platforms, mobile apps, customer reviews, and IoT devices**. When this data is stored in different systems, it becomes difficult to understand sales patterns, customer behavior, and inventory requirements.

A unified big data architecture can integrate all these data sources into a common platform.

Proposed Architecture

POS Systems + E-Commerce Websites

- Mobile Applications + Warehouses
- Customer Reviews + IoT Sensors**

↓

Data Collection & Ingestion – Kafka / APIs / ETL

↓

Distributed Cloud Storage / Data Lake

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Data Cleaning & Processing – Spark / Hive

↓

Unified Data Warehouse

↓

Analytics & Machine Learning

↓

Demand Forecasting + Inventory Optimization + Customer Insights + Business Dashboards

Apache Kafka can collect sales and customer events continuously from different locations. The **cloud data lake** stores large amounts of structured and unstructured data such as transaction records, product information, and customer reviews.

Apache Spark processes the collected data and identifies useful patterns. The processed information can be stored in a **data warehouse** for reporting and analysis.

Machine learning models can predict **product demand, customer preferences, and inventory requirements**. Business dashboards can then provide managers with real-time information about sales, stock levels, and customer trends.

This architecture gives the organization a **single and integrated view of its global operations**, helping it reduce inventory problems, improve customer service, increase sales, and make faster business decisions.

This architecture provides a complete view of the customer and allows the company to deliver personalized recommendations immediately.

2. Distributed Computing for Scientific Research Data

Scientific research can generate extremely large datasets from simulations, satellites, laboratories, sensors, and experiments. Processing this data on a single computer can take a very long time.

Distributed computing divides the workload among multiple computers.

Centralized vs Distributed Processing

In a centralized system, one powerful computer performs most of the computation. As the dataset becomes larger, processing time and hardware requirements increase.

In a distributed system, the dataset is divided into smaller parts and processed simultaneously by multiple machines.

For example, **Apache Spark** can divide a large scientific dataset into partitions and process them in parallel.

Advantages

- Faster processing

- Better scalability
- Parallel execution
- Fault tolerance
- Ability to process very large datasets
- Efficient use of computing resources

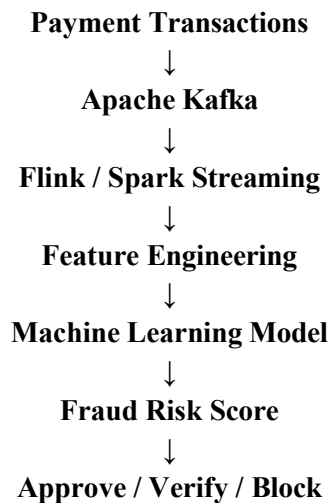
If one machine processes 1 billion records, the operation may take a long time. If the work is divided among 100 machines, many records can be processed simultaneously.

Therefore, distributed computing is more suitable for large-scale scientific research because it improves both computational efficiency and scalability.

3. Fraud Detection Framework for a Digital Payment Platform

Digital payment systems process a large number of transactions every second. Traditional fraud detection methods may not be fast enough to identify suspicious transactions immediately.

Proposed Framework



Kafka can collect transactions continuously. A stream-processing framework can analyze each transaction as it arrives.

The system can consider:

- Transaction amount
- Location
- Device
- Transaction frequency
- Customer spending pattern
- Merchant

- Previous fraudulent activity

Machine learning models such as **Random Forest, XGBoost, Logistic Regression, and anomaly detection algorithms** can identify unusual behavior.

For example, if a customer's account normally makes small transactions in one region but suddenly performs several large transactions from another region, the system can assign a high fraud score.

Streaming processing provides speed, while predictive models provide better detection accuracy. Together, they reduce financial losses and improve payment security.

4. Multimedia Data in Big Data Environments

Multimedia data includes videos, audio recordings, images, and other large files. These datasets create several challenges because they are much larger than normal text or structured records.

Major Challenges

1. **Large storage requirements** – High-resolution videos and images consume significant storage.
2. **High processing requirements** – Multimedia analysis requires considerable CPU or GPU power.
3. **Different formats** – Data can exist in formats such as MP4, JPEG, PNG, WAV, and others.
4. **Fast data generation** – Cameras and streaming systems continuously generate new data.
5. **Metadata management** – Information such as location, time, device, and content type must also be stored.

Suitable Technologies

Cloud object storage such as **Amazon S3, Azure Blob Storage, or Google Cloud Storage** is suitable for large multimedia files.

Hadoop HDFS can also be used for distributed storage.

For processing, **Apache Spark** can perform large-scale data processing, while **GPU-based systems** can accelerate image and video analysis.

Metadata can be stored in databases such as PostgreSQL or NoSQL systems.

Compression and lifecycle policies should also be used to reduce storage costs.

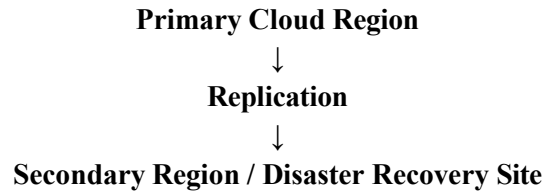
Therefore, a combination of distributed storage, cloud object storage, metadata databases, compression, and parallel processing provides an efficient multimedia big data solution.

5. Disaster Recovery and Business Continuity for a Cloud Big Data Platform

A cloud-based big data platform may contain critical business information. Hardware failures, software errors, cyberattacks, or natural disasters can interrupt operations.

A disaster recovery strategy should ensure that data and services can be restored quickly.

Proposed Strategy



Important data should be replicated to another availability zone or region.

Key Techniques

Redundancy: Multiple servers and storage systems prevent a single hardware failure from stopping the platform.

Replication: Data is copied to another location so that it remains available if the primary system fails.

Automated Failover: If the primary service becomes unavailable, traffic can automatically be redirected to the backup system.

Regular Backups: Periodic backups protect against accidental deletion and corruption.

Monitoring: Automated monitoring can detect failures and trigger recovery procedures.

Business Continuity

The organization should define:

- **RTO:** Maximum acceptable time to restore the service.
- **RPO:** Maximum acceptable amount of data loss.

Regular disaster recovery testing is also necessary.

This approach provides high availability and ensures that business operations can continue even after major failures.

6. Data Quality Problems in Large-Scale Analytics

Data quality has a direct effect on the reliability of analytics. Poor-quality data can produce incorrect reports and misleading machine learning predictions.

Major Data Quality Problems

Inconsistency: The same information may be stored differently in different systems.

Duplication: The same customer or transaction may appear multiple times.

Missing Values: Important fields may be empty or unavailable.

For example, if customer information is duplicated, the company may incorrectly calculate the number of customers.

Data Governance Strategy

A proper governance framework should include:

- Data quality rules

- Data validation
- Duplicate detection
- Standard data formats
- Metadata management
- Data ownership
- Access control
- Data lineage
- Regular quality monitoring

ETL pipelines can clean and transform data before it enters the analytics system.

A data catalog can document where data comes from and how it is used.

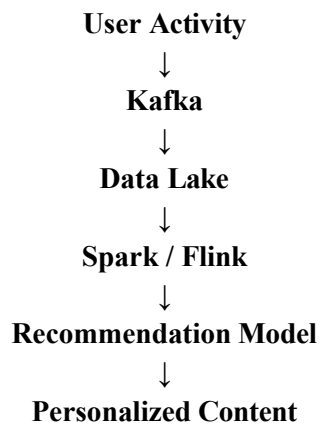
Data quality dashboards can continuously monitor missing values, duplicates, and inconsistent records.

Therefore, strong data governance ensures that analytics systems use accurate, consistent, and trustworthy data.

7. Scalable Recommendation System for an Online Streaming Service

An online streaming platform such as a video or music service collects information about what users watch or listen to. This information can be used to provide personalized recommendations.

Proposed Architecture



The system can collect:

- Movies or songs viewed
- Watch time
- Search history
- Likes and ratings
- Skipped content

- Recently watched items

Kafka can process user events in real time. Spark can analyze historical user behavior.

Recommendation Algorithms

Collaborative filtering recommends content based on similar users.

Content-based filtering recommends content with characteristics similar to previously consumed items.

A hybrid recommendation system can combine both methods.

Distributed processing allows millions of users and large amounts of behavior data to be analyzed efficiently.

For example, if users who watch a particular movie frequently watch another movie, the system can recommend the second movie to similar users.

Better recommendations increase user engagement, viewing time, and customer satisfaction.

8. Role of Cloud Computing in Big Data

Cloud computing has become an important part of modern big data systems because organizations can access large amounts of storage and computing power without maintaining all the physical hardware themselves.

Advantages

Scalability: Resources can be increased or decreased according to workload.

Cost Management: Organizations can avoid large investments in physical infrastructure and pay according to resource usage.

Flexibility: Different storage, database, analytics, and machine learning services can be selected according to requirements.

Availability: Cloud providers offer backup, replication, and high-availability features.

Faster Deployment: Big data platforms can be deployed much faster than traditional data centers.

Limitations

Cost Control: Poor resource management can result in unexpectedly high bills.

Security Risks: Sensitive data must be properly protected using encryption and access controls.

Vendor Lock-in: Moving applications from one cloud provider to another may be difficult.

Network Dependency: Cloud applications depend heavily on reliable network connectivity.

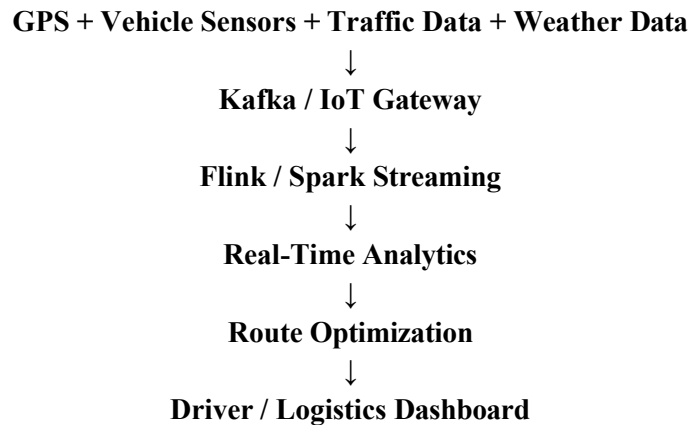
Compliance: Organizations must ensure that data storage and processing comply with applicable regulations.

Overall, cloud computing provides excellent scalability and flexibility, but organizations need proper cost management, security, and governance.

9. Real-Time Analytics for Transportation and Logistics

Transportation companies generate data from GPS devices, vehicles, traffic systems, warehouses, and delivery applications. Real-time analytics can improve routing and operational decisions.

Proposed Architecture



GPS data can be continuously collected from vehicles.

A stream-processing system can combine vehicle location with traffic and weather information.

Route Optimization

Machine learning and optimization algorithms can identify better routes based on:

- Traffic congestion
- Distance
- Delivery priority
- Fuel consumption
- Weather
- Road conditions

If a road suddenly becomes congested, the system can recommend an alternative route.

Benefits

- Reduced travel time
- Lower fuel consumption
- Improved delivery schedules
- Better fleet utilization
- Faster response to traffic problems
- Improved customer service

Real-time dashboards can help managers monitor vehicles, deliveries, delays, and fuel usage.

10. Ethical and Legal Challenges of Big Data

Big data provides many benefits, but collecting and analyzing large amounts of personal information also creates ethical and legal concerns.

Major Challenges

Privacy: Organizations may collect information that users do not expect to be collected.

Lack of Consent: Data should not be used for purposes that users have not agreed to.

Data Security: Poorly protected data can be exposed through cyberattacks.

Bias: Machine learning models can produce unfair results if their training data contains bias.

Lack of Transparency: Users may not understand how automated decisions are made.

Data Retention: Organizations should not keep personal information longer than necessary.

Responsible Data Policies

Organizations should:

- Obtain appropriate user consent
- Collect only necessary information
- Encrypt sensitive data
- Apply strong access controls
- Anonymize data where possible
- Maintain audit logs
- Regularly test machine learning models for bias
- Clearly explain how data is being used
- Provide mechanisms for correcting inaccurate information
- Follow applicable privacy and data-protection laws
- Establish clear data retention and deletion policies

Importance of Trust

Responsible data management improves customer confidence. Organizations that protect personal information and use it transparently are more likely to maintain long-term user trust.

Therefore, big data should not be evaluated only on its technical capabilities. Privacy, fairness, security, legal compliance, and ethical responsibility must also be considered when designing a big data system.